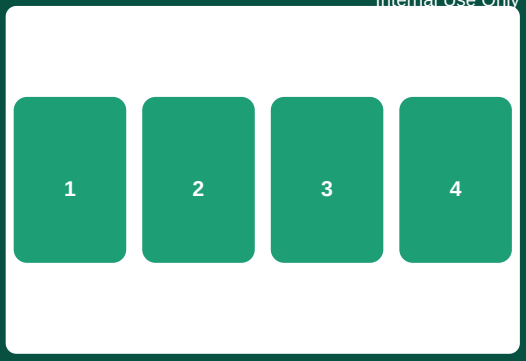
QUALITY & PROCESS

Box & Whisker Plot

Visualise the distribution of a dataset — median, quartiles, spread, and outliers — in a compact graphical format for



WHEN TO USE

Comparing distributions across groups — shifts, machines, operators, suppliers, time periods, or product lines.

HOW TO APPLY — STEPS

1. Collect data for each comparison group
2. Calculate median, Q1, Q3, and whisker bounds
3. Plot and compare distributions visually side by side
4. Identify which groups have the widest spread or most outliers
5. Investigate root causes of differences between groups

PRACTITIONER TIP

! A wide box means high variation — usually more important to address than a shifted mean. Reduce spread first.

KEY OUTPUTS

- Box plot chart
- Summary statistics table
- Comparison insights and hypotheses

HOW TO MEASURE SUCCESS

Interquartile range reduction; outlier frequency reduction; between-group variation reduction

TOOL CATEGORY

Quality & Process